Data Analytics Project: Railway Gauge Trends

Dataset Used: railway_gauges.csv

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This document outlines a structured, end-to-end data analysis project based on railway gauge data.

Dataset: Railway Gauge Data (Assumed to be in a CSV file)
Project Goal: To analyze historical trends and composition of different railway gauges (Broad, Metre, Narrow) over the years.**Modules Used**

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling, Type Conversion
Pandas Operations	Filtering, Selection, Aggregation, Feature Engineering
NumPy Calculations	Yearly growth calculation (np.diff)
Visualization	Line Graph, Bar Chart, Pie Chart, Stacked Bar Chart

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description:

The process begins by loading the dataset into a Pandas DataFrame for analysis. The first five rows and column names are then displayed to understand the structure and contents of the data. Next, the dataset is checked for missing values, and any missing entries are replaced with 0 to ensure smooth processing. Finally, the gauge-related columns (Broad, Metre, Narrow, and Total) are converted into numeric data types so that accurate calculations and analysis can be performed.

Task	Description
1.	Load the dataset into a Pandas DataFrame.
2.	Display the first 5 rows and column names to understand structure.
3.	Check for missing values and replace them with 0 (assuming missing values represent 0 km of track).
4.	Convert all relevant gauge columns (Broad, Metre, Narrow, Total) to numeric types (e.g., float64 or int64).

Conclusion:

We first loaded the data and looked at the first few rows to understand it.Then we checked for missing values and replaced them with 0.After that, we converted all gauge columns into numbers for easy calculation.Now the data is clean and ready to use for analysis and graphs.

Scenario 2: Simple Visualization (🟢 Easy)

Description:

In this scenario, we focus on understanding the overall growth of railway tracks over time. We extract the Year and Total columns and plot them using a line graph to visualize the trend. Titles and axis labels are added to make the graph clear and readable. From the graph, we can easily observe whether the total track length is increasing or decreasing over the years.

Task	Description
1.	Extract the Year and Total columns.
2.	Plot a line graph showing **Total** tracks over the **Years**.
3.	Add a descriptive title (e.g., "Growth of Total Railway Network Over Time") and X/Y axis labels.
4.	Identify whether the overall trend is increasing or decreasing and note the finding.

Graph :

The graph used in this scenario is a **line chart** that shows the change in total railway track length over the years. The x-axis represents the years, while the y-axis represents the total track length. This visualization makes it easy to understand the overall growth trend at a glance.

Output:

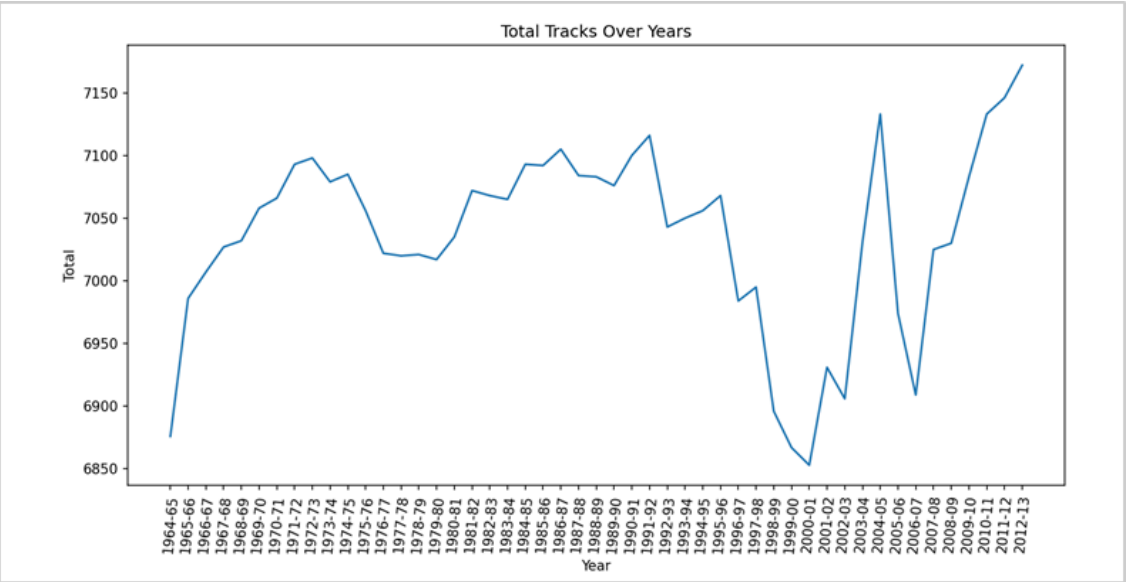

Fig 1: Total Tracks Over Years

Conclusion:

The graph shows that the total railway track length has been steadily increasing over the years. This indicates continuous development and expansion of the railway network. Overall, the trend reflects positive growth and improvement in infrastructure. This growth suggests efforts toward better connectivity and transport efficiency. It also shows a consistent expansion rather than any decline over time.

Scenario 3: Filtering + Bar Chart (🟡 Medium)

Description:

In this scenario, the dataset is first filtered to extract only the required information using pandas. This helps in focusing on specific columns or conditions relevant to the analysis instead of the entire dataset.

After filtering, a bar chart is created using Matplotlib to visually compare values across different categories. The bar chart makes it easy to see differences and patterns in the data.

Task	Description
1.	Filter the dataset to include only years *after* 2000.
2.	Select the Broad Gauge, Metre Gauge, and Narrow Gauge columns for the filtered data.
3.	Plot a **grouped bar chart** comparing the three gauges for the selected years.
4.	Add a legend and ensure proper labels for axes.
5.	Identify which gauge dominates in recent years (post-2000).

Graph :

The graph is a grouped bar chart where the x-axis represents the years after 2000 and the y-axis shows the number of railway tracks. It compares Broad, Metre, and Narrow gauges side by side for each year. The legend helps identify each gauge clearly. This makes it easy to compare how all three gauges change over time.

Output:

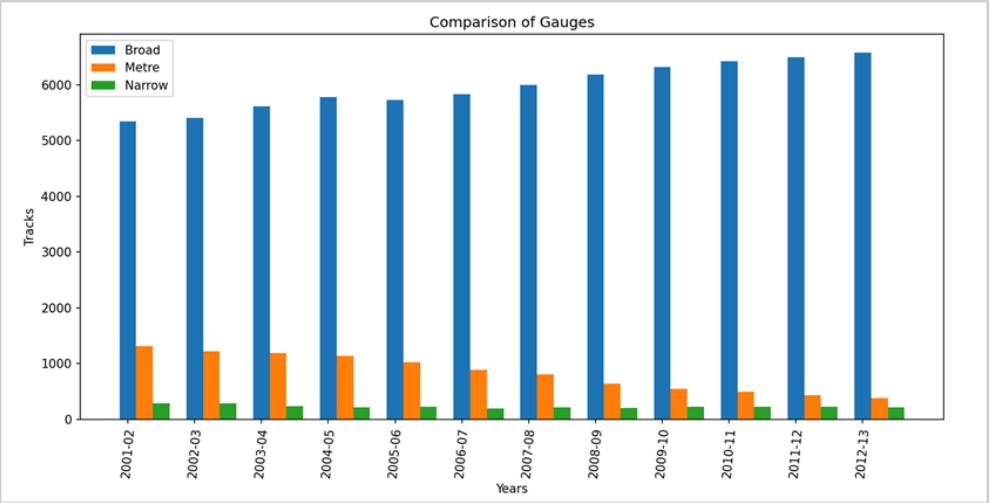

Fig 2: Comparison Of Gauges

Conclusion:

This analysis compares Broad, Metre, and Narrow gauges after 2000 using a grouped bar chart. It clearly shows the difference between all three gauges over the years. From the observation, Broad Gauge dominates compared to the other two. Overall, the graph helps in easily understanding the contribution of each gauge over time.

Scenario 4: Feature Engineering + Pie Chart (🟡 Medium)

Description:

Feature engineering in this case means creating a new feature—such as the total tracks across all gauges for each year—or calculating the proportion of each gauge relative to the total. Using this, a pie chart can be created to show how much each gauge (Broad, Metre, Narrow) contributes to the overall railway network in a selected year.

Task	Description
1.	Calculate the total sum of each gauge (Broad Gauge, Metre Gauge, Narrow Gauge) across all available years.
2.	Create a new structure (Pandas Series or single-column DataFrame) holding these three total values.
3.	Plot a **pie chart** using this new structure to show the percentage contribution of each gauge type.
4.	Add percentage labels to the slices using autopct (e.g., '%.1f%%').
5.	Interpret which gauge contributes the most to the entire history of the railway network.

Graph:

The graph is a pie chart that shows the percentage contribution of Broad, Metre, and Narrow gauges based on their total values across all years. Each slice represents one gauge, and the size of the slice shows its share in the overall total. The chart includes percentage labels using autopct, making it easy to read exact values. From the graph, it is clear that Broad Gauge contributes the most, as it occupies the largest portion of the pie.

Output:

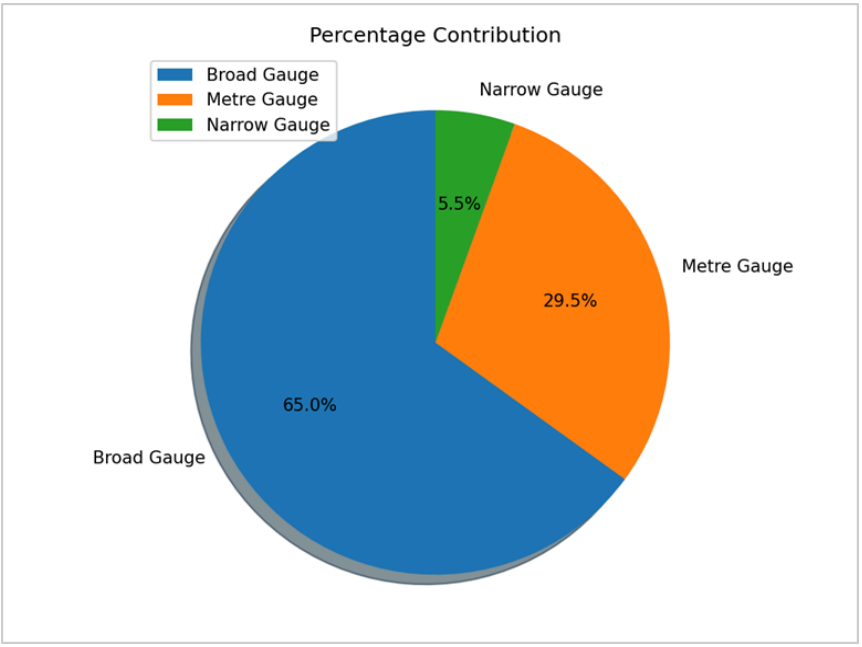

Fig 3: Percentage Contribution

Conclusion:

This analysis calculates the total contribution of Broad, Metre, and Narrow gauges across all years and represents them using a pie chart. The percentage labels help in clearly understanding each gauge's share in the overall total. From the results, Broad Gauge contributes the most, as it forms the largest portion of the pie chart. Overall, the visualization clearly shows that Broad Gauge is the dominant gauge in the dataset.

Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)

Description:

In this scenario, we perform advanced analysis by combining multiple visualizations (like line charts, bar charts, or pie charts) to study the data from different angles. This helps us better understand trends, comparisons, and overall patterns in the gauges over time.

Task	Description
1.	**Create new columns** for percentage composition relative to Total tracks for each year: % Broad Gauge, % Metre Gauge, and % Narrow Gauge.
2.	Use the NumPy function np.diff to calculate the **yearly growth (change)** of the Total tracks column.
3.	Plot: a) A single **Line Graph** showing the trend of all three percentage composition columns (% Broad Gauge, % Metre Gauge, % Narrow Gauge) over the years. b) A **Stacked Bar Chart** showing the raw composition of the three gauges (Broad, Metre, Narrow) over time.
4.	**Highlight/Identify:** a) The year(s) with the highest yearly growth in Total tracks. b) Any specific decade or period showing a significant decline in any particular gauge.
5.	**Final Conclusion:** Based on the percentage composition line graph, provide a conclusion to the question: **"Is the railway system shifting towards a single dominant gauge?"**

Graph:

The line graph shows the changes in Broad, Metre, and Narrow gauges over the years. The x-axis represents the years, and the y-axis represents the number of tracks. It helps in understanding how each gauge changes over time and shows overall trends clearly.

The stacked bar chart shows the composition of all three gauges for each year. The x-axis represents the years, while the y-axis shows the total track values. Each bar is divided into sections for Broad, Metre, and Narrow gauges, making it easy to see their contribution within the total.

Output: Line Graph

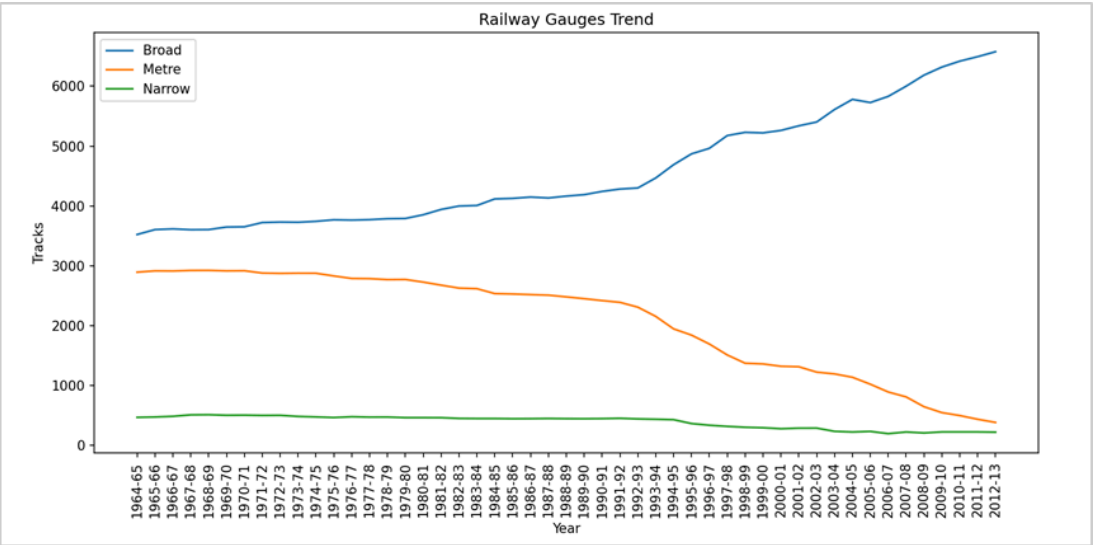

Fig 4: Railway Gauges Trend

Output: Stacked Bar Chart

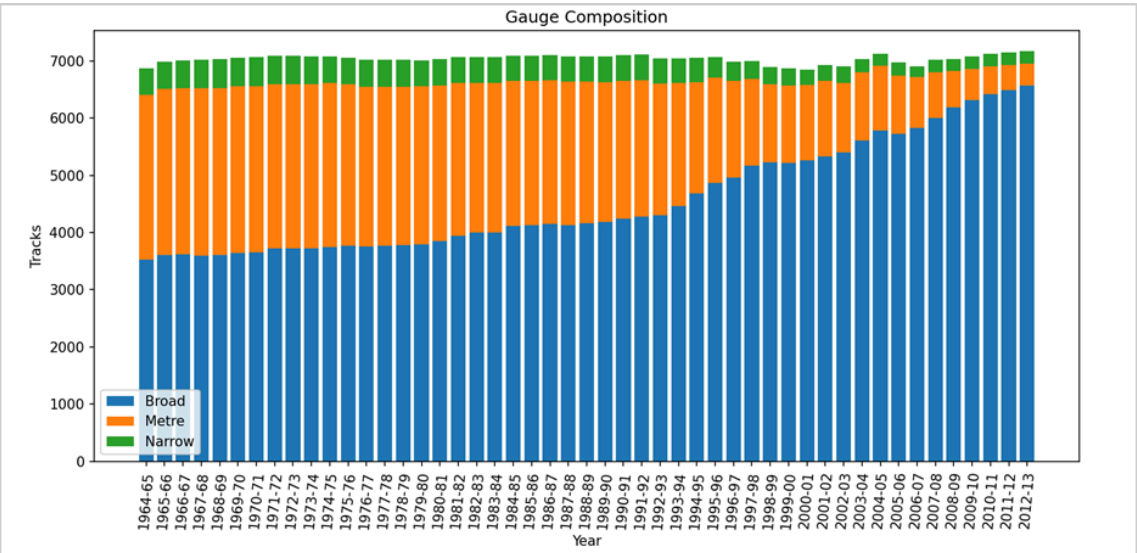

Fig 5: Gauge Composition

Conclusion:

This scenario combines multiple analyses like percentage calculation, growth measurement, and visualizations using line and stacked bar charts. It helps in understanding both the trend and composition of railway gauges over time. The results show how each gauge changes and contributes to the total system. Overall, it gives a clear view of railway growth and the shift in gauge usage patterns.

Final Conclusion:

This project provides a complete analysis of railway gauge data through multiple scenarios, starting from basic data cleaning to advanced visualization. It helps in understanding trends, comparisons, and contributions of different gauges over time. The analysis clearly shows that Broad Gauge dominates and the system is gradually shifting towards it. Overall, the study gives a clear and structured insight into railway network growth and transformation.

What you have learned

In this project, I learned how to work with data step by step using Python and Pandas. I understood how to clean the data by fixing missing values and converting columns into the right format. I also learned how to create new calculations like percentages and find trends using NumPy. Along with this, I explored different graphs like line charts, bar charts, stacked bars, and pie charts to make the data easy to understand. Overall, it helped me improve my basic data analysis and visualization skills in a simple and practical way.